

Modern Multi-Object Tracking Systems

A Comparative Study of SORT, DeepSORT, ByteTrack, and BoT-SORT

1. Introduction

Multi-object tracking (MOT) assigns a consistent identity to every object in a video as it moves, is occluded, and re-appears. Nearly all modern trackers follow the tracking-by-detection paradigm: an object detector (e.g. YOLO, Faster R-CNN) locates objects in each frame, and a separate tracking module predicts motion and associates new detections with existing tracks. The four systems below represent successive stages in this paradigm's evolution, each addressing a specific weakness of its predecessor.

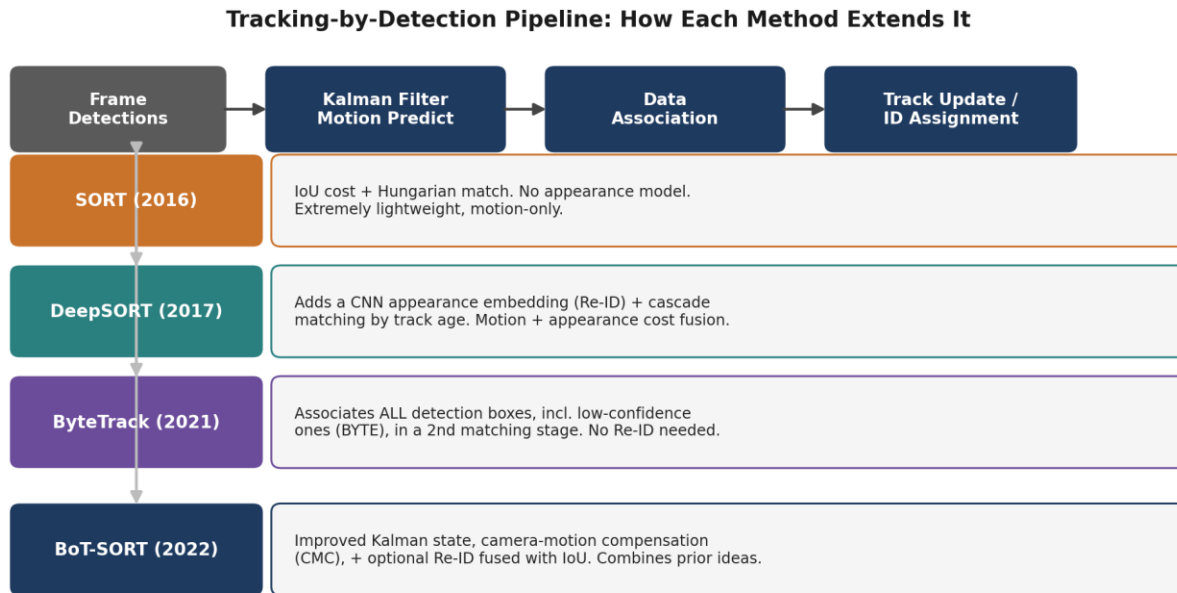


Figure 1. Shared tracking-by-detection pipeline, with the key modification each method introduces.

2. Method Overviews

2.1 SORT (Simple Online and Realtime Tracking, 2016)

SORT associates detections purely through motion: a Kalman filter predicts each track's next bounding box, and the Hungarian algorithm matches predictions to new detections using Intersection-over-Union (IoU) cost. It carries no appearance model, which keeps it extremely fast but fragile: once a track is occluded for a few frames, its identity is easily lost or swapped with a nearby object.

2.2 DeepSORT (2017)

DeepSORT extends SORT with a CNN-based appearance descriptor (a small re-identification, or Re-ID, network) trained on person images. Matching combines a Mahalanobis motion-distance gate with cosine distance between appearance embeddings, and uses cascade matching that prioritises recently-seen tracks. This substantially reduces identity switches through short occlusions, at the cost of an extra network pass per detected box.

2.3 ByteTrack (2021)

ByteTrack's key insight is that conventional trackers discard low-confidence detection boxes (often partially-occluded objects), throwing away useful evidence. Its BYTE association matches high-score boxes first, then makes a second association pass between remaining tracks and the low-score boxes, recovering occluded objects without needing an appearance network at all. This yields large accuracy gains in crowded scenes while remaining real-time.

2.4 BoT-SORT (2022)

BoT-SORT ("Bag of Tricks" SORT) combines ideas from all three predecessors and adds two of its own: an improved Kalman filter state vector for more accurate motion prediction, and camera-motion compensation (CMC), which corrects for a moving

camera using image registration before computing motion cost. An optional Re-ID branch fuses IoU and appearance distance, similar in spirit to DeepSORT but built on ByteTrack's stronger association backbone.

3. Comparative Analysis

Table 1 summarises the trackers along the three requested dimensions. Figures reflect each method's original paper (mixed MOT16/MOT17 protocols) and later replications; absolute numbers vary with the detector and hardware used, so they should be read as relative indicators rather than exact, hardware-independent constants.

Method	Accuracy (MOT17-style)	Speed	Best-fit Applications
SORT	Low (MOTA ~60, IDF1 ~45). Frequent ID switches under occlusion.	Very high (~250+ FPS, excl. detector) - pure Kalman + IoU.	Low-latency embedded use cases where identity switches are tolerable, e.g. simple people-counting, prototyping.
DeepSORT	Moderate (MOTA ~61, IDF1 ~62). Fewer ID switches via appearance cues.	Medium (~40 FPS) - extra CNN embedding pass per box.	Re-identification-sensitive tasks: retail analytics, surveillance across brief occlusions.
ByteTrack	High (MOTA 80.3, IDF1 77.3, HOTA 63.1 on MOT17 test).	Real-time (~30 FPS incl. detector on a V100) - no ReID network needed.	Crowded-scene tracking: sports analytics, traffic/pedestrian counting, general-purpose real-time MOT.
BoT-SORT	Highest (MOTA 80.5, IDF1 80.2, HOTA 65.0 on MOT17 test).	Lower with ReID+CMC enabled; near real-time without it.	Applications needing robust re-ID under camera motion: drone/PTZ surveillance, autonomous driving perception.

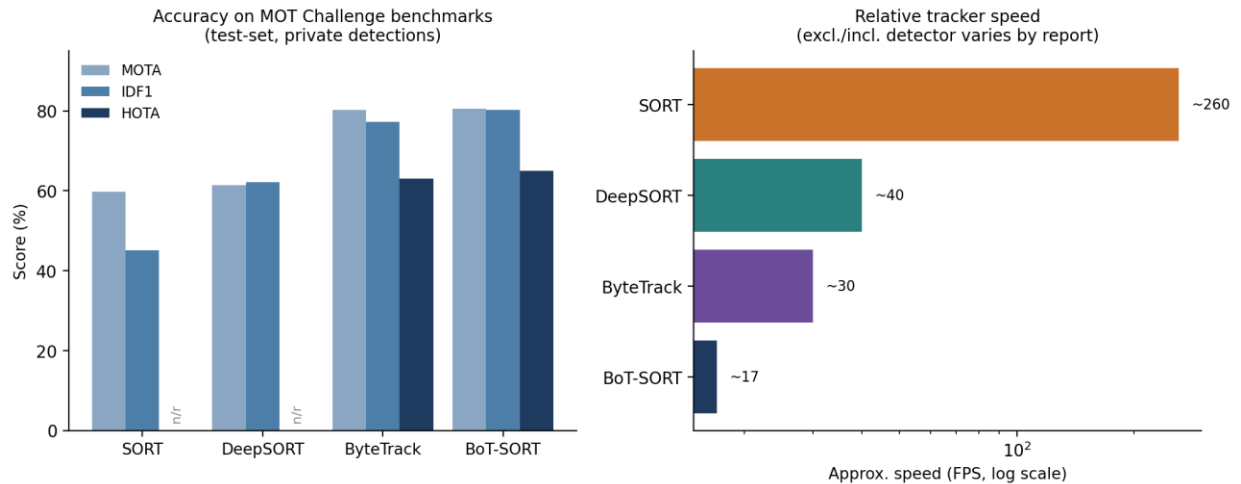


Figure 2. Left: accuracy on MOT-Challenge test sets (HOTA not reported for SORT/DeepSORT, predating the metric). Right: approximate relative speed; exact FPS depends heavily on GPU and detector choice.

3.1 Accuracy

Accuracy rises sharply with each generation. SORT and DeepSORT plateau in the 60s (MOTA) because their motion models and/or appearance networks still lose objects in dense, fast-moving scenes. ByteTrack's low-score recovery and BoT-SORT's

motion/appearance refinements push MOTA above 80 and HOTA into the mid-60s on MOT17, the current state-of-the-art range for tracking-by-detection systems.

3.2 Speed

Speed trends in the opposite direction. SORT's pure geometric matching is the fastest by a wide margin. DeepSORT's appearance CNN adds meaningful latency. ByteTrack recovers much of that speed by avoiding a Re-ID network altogether, remaining real-time even on modest hardware. BoT-SORT is the most accurate but also the heaviest when its optional Re-ID and CMC modules are enabled, though it can be run in a lighter, ByteTrack-like mode.

3.3 Applications

The right tracker depends on the constraint that matters most. SORT suits latency-critical, low-power deployments where occasional ID switches are acceptable. DeepSORT fits tasks needing re-identification across brief occlusions, such as retail footfall analysis. ByteTrack is a strong general-purpose default for crowded real-time scenes (sports analytics, pedestrian/vehicle counting). BoT-SORT is preferable when the camera itself moves (drones, PTZ cameras, dashcams) and maximum identity-preservation accuracy is worth the extra compute.

4. Conclusion

SORT, DeepSORT, ByteTrack, and BoT-SORT illustrate a clear design trajectory in multi-object tracking: from motion-only association, to appearance-aware matching, to smarter use of all available detections, to combining every prior trick with camera-motion awareness. In practice, the choice is a trade-off between accuracy, compute budget, and scene characteristics (crowd density, camera motion) rather than a single “best” tracker.

References

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